

CLASS 11-CBSE MATHEMATICS
CHAPTER: BINOMIAL THEOREM-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions [1X3 = 3 MARKS]

1. Find $x^5 + 10x^4a + 40x^3a^2 + 80x^2a^3 + 80xa^4 + 32a^5$
2. Find the term independent of x in the expansion of $\left(\sqrt{\frac{x}{3}} + \frac{3}{2x^2}\right)^{10}$
3. Find the coefficient of x^5 in the expansion of $(1+x^2)^5(1+x)^4$.

SECTION B

Q.II Answer the following questions [2X5 = 10 MARKS]

4. If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a+b)^{n+3}$ are equal, then find n .
5. If the term free from the expansion of $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ is 405, find the value of k .
6. Using Binomial Theorem, evaluate $(96)^3$.
7. Find the 6th term in expansion of $\left(2x^2 - \frac{1}{3x^2}\right)^{10}$
8. Find the middle term in the expansion of
 - a) $\left(\frac{x}{a} - \frac{a}{x}\right)^{10}$
 - b) $\left(3x - \frac{x^3}{6}\right)^9$

SECTION C

Q.III Answer the following questions [3X3 = 9 MARKS]

9. Show that $9^{n+1} - 8n - 9$ is divisible by 64, whenever n is a positive integer.



10. Find the middle terms in the expansions of $\left(3 - \frac{x^3}{6}\right)^7$
11. If a and b are distinct integers, prove that $a - b$ is a factor of $a^n - b^n$, whenever n is a positive integer.

SECTION D

Q.IV Answer the following questions [4X2 = 8 MARKS]

12. In the expansion of $(1 + a)^{m+n}$, prove that coefficients of a^m and a^n are equal.
13. Expand using Binomial Theorem $\left(1 + \frac{x}{2} - \frac{2}{x}\right)^4$. $x \neq 0$.

